Lesson 9.4: Equilibrium and solutions

Game theory

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Uncertainty in games

	b_1	b_2
a_1	2	0
a_2	1	3

	b_1	b_2	min
a_1	2	0	0
a_2	1	3	1
Max	2	3	

- this game doesn't have a unique solution/value
- does it have a solution/value at all?
- playing *Maximin* strategies places bounds
  - row player guaranteed at least 1 by playing a_2
  - column player can ensure at most 2 by playing b_1
- what is a fair value for this game?

Patterns of play

	b_1	b_2
a_1	2	0
a_2	1	3

	A	B	payoff
1	a_2	b_1	1
2	a_1	b_1	2
3	a_1	b_2	0
4	a_2	b_2	3
5	a_2	b_1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- average payoff 1.5
- what should they play on their next move?
- predictable pattern

Mixed strategies

Random mixtures:

	b_1	b_2	$\frac{1}{2}b_1\frac{1}{2}b_2$	min
a_1	2	0	1.0	0
a_2	1	3	2.0	1
$\frac{1}{2}a_1\frac{1}{2}a_2$	1.5	1.5	1.5	1.5
Max	2	3	2.0	

- discourage a_2 by playing b_1 more frequently?
- how much more frequently?